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(54) **OUTLET HEAT SINK FOR COOLING
SYSTEM OF A SUPERCOMPUTER
ELECTRONIC BOARD**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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atomique et aux énergies alternatives**
(CEA), Paris (FR)

10,481,652 B2 * 11/2019 Rice H05K 7/20927
10,739,084 B2 * 8/2020 Tsai H05K 7/20254
11,157,050 B1 * 10/2021 Lunsman H05K 7/20509
(Continued)

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FOREIGN PATENT DOCUMENTS

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CN 112015250 A * 12/2020 G06F 1/183
EP 2770810 B1 2/2014
(Continued)

OTHER PUBLICATIONS

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patent is extended or adjusted under 35
U.S.C. 154(b) by 146 days.

Zhao Xi, "Server integral parallel flow cold plate liquid cooling
system", Dec. 1, 2020, Chengdu Longwei System Tech Co Ltd,
Entire Document (Translation of CN 112015250). (Year: 2020).
(Continued)

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(57) **ABSTRACT**

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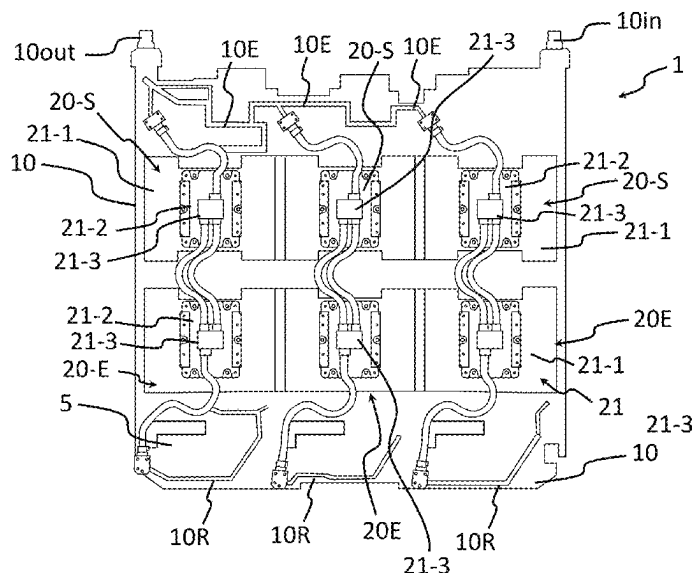
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CPC **H05K 1/021** (2013.01); **H05K 2201/064**
(2013.01); **H05K 2201/066** (2013.01)

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An outlet heat sink for a liquid cooling system for an
electronic board, including a cold inlet connector, a hot inlet
connector and a hot outlet connector. An upper part of a
cooling block, on the one hand, conveys the flow of heat
transfer fluid entering through the cold inlet connector to the
lower part of the cooling block so as to collect heat gener-
ated by the electronic board by being conveyed to the hot
outlet connector, and on the other hand, conveys the flow of
heated-up heat transfer fluid received on the hot inlet con-
nector directly to the hot outlet connector.

See application file for complete search history.

10 Claims, 6 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

11,452,237	B2 *	9/2022	Tan	H05K 7/20272
2005/0128705	A1 *	6/2005	Chu	H01L 23/473
				361/689
2008/0026509	A1 *	1/2008	Campbell	G06F 1/20
				165/80.4
2014/0233175	A1 *	8/2014	Demange	H05K 7/20254
				361/699
2016/0118317	A1	4/2016	Shedd et al.	
2018/0340744	A1	11/2018	Tsai et al.	
2019/0387609	A1 *	12/2019	Chen	H01L 23/473
2021/0076539	A1 *	3/2021	Raeth	H05K 7/20254
2021/0307195	A1 *	9/2021	Tian	F28F 3/044
2023/0025167	A1 *	1/2023	Gao	H05K 7/20254
2024/0074100	A1 *	2/2024	Saksager	G06F 30/17

FOREIGN PATENT DOCUMENTS

EP		2770809	B1	4/2019
EP		3500079	A1	6/2019
WO		2019115963	A1	6/2019
WO	WO-2022192031	A1 *	9/2022	 H05K 1/0203

OTHER PUBLICATIONS

Search Report issued in EP22306292.8 on Feb. 10, 2023 (5 pages).

* cited by examiner

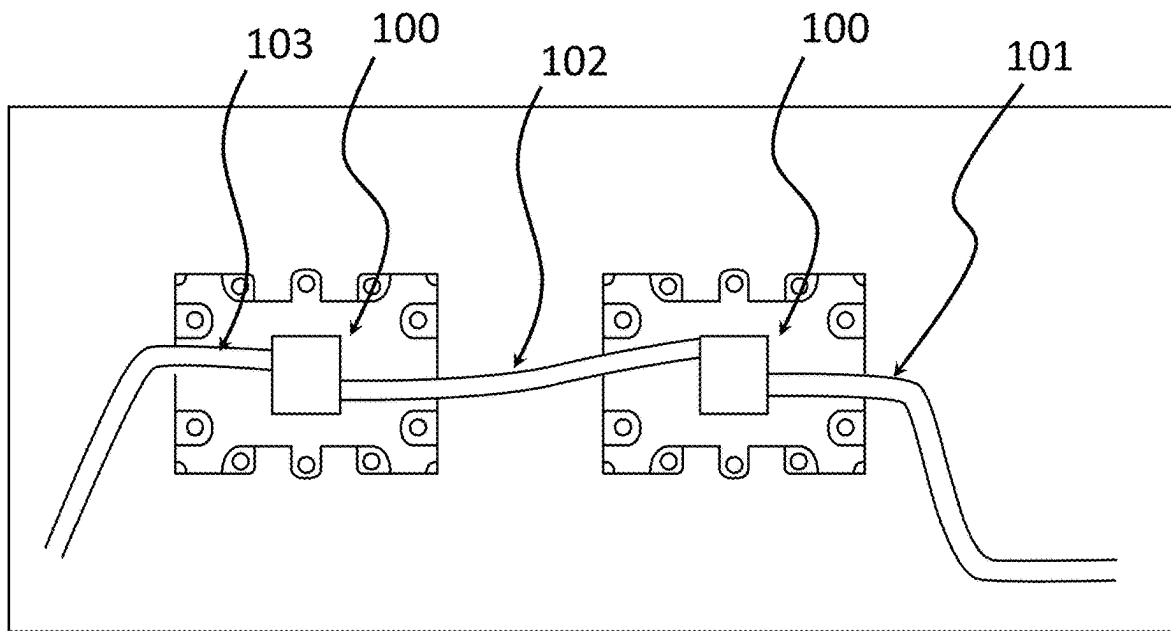


FIG. 1

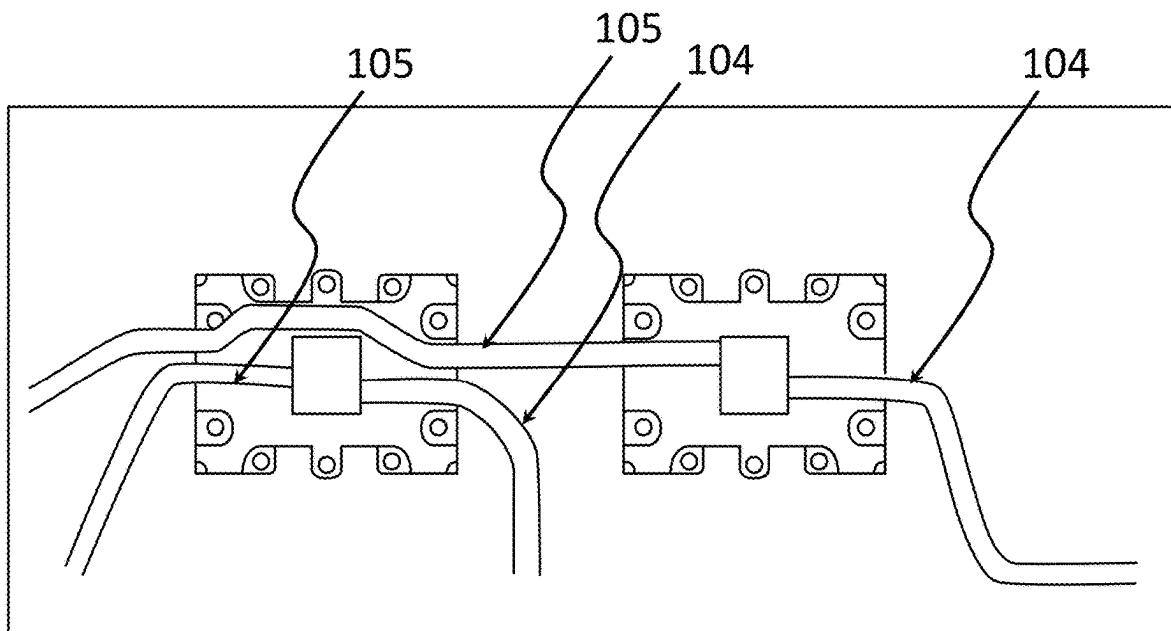


FIG. 2

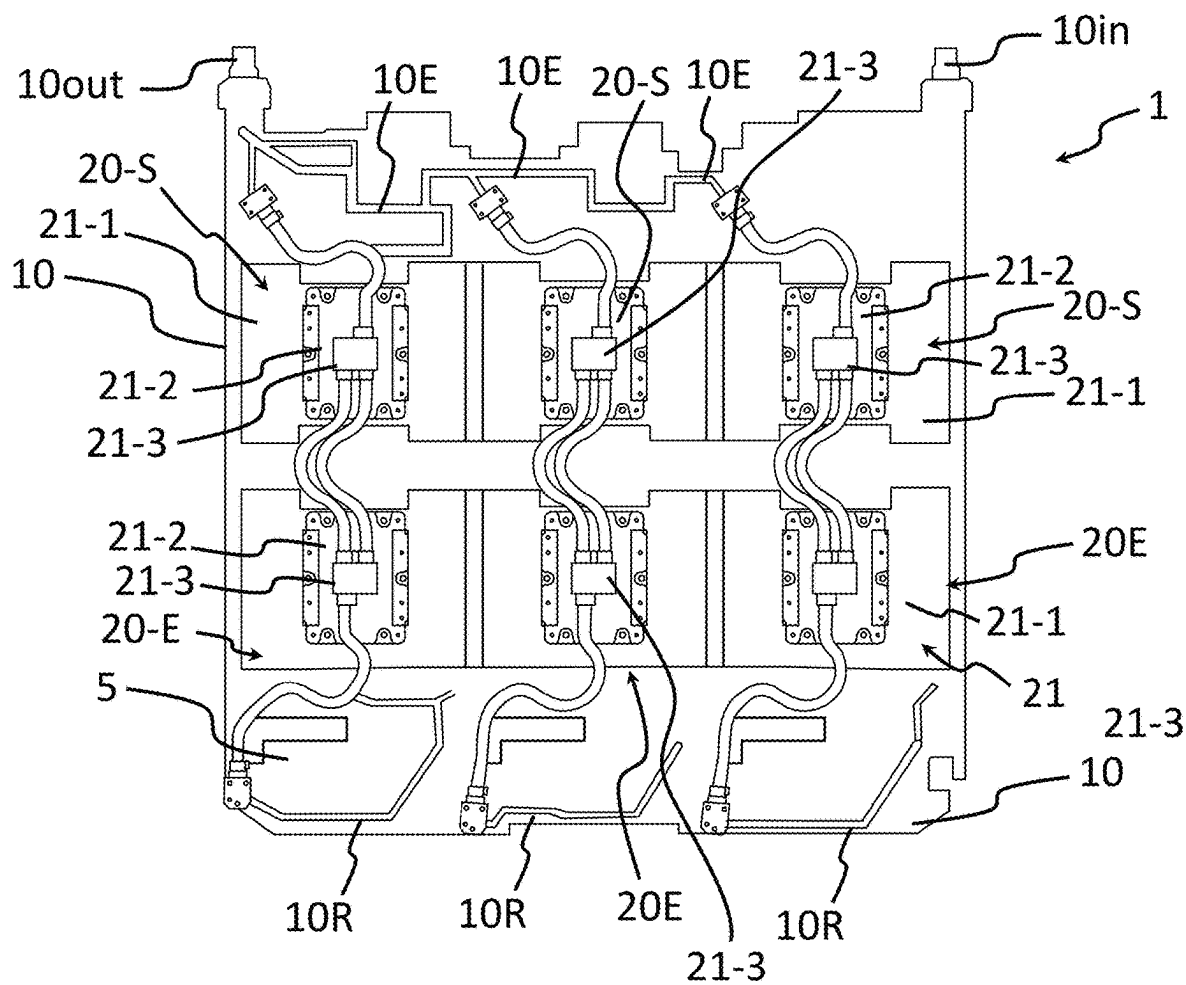


FIG. 3

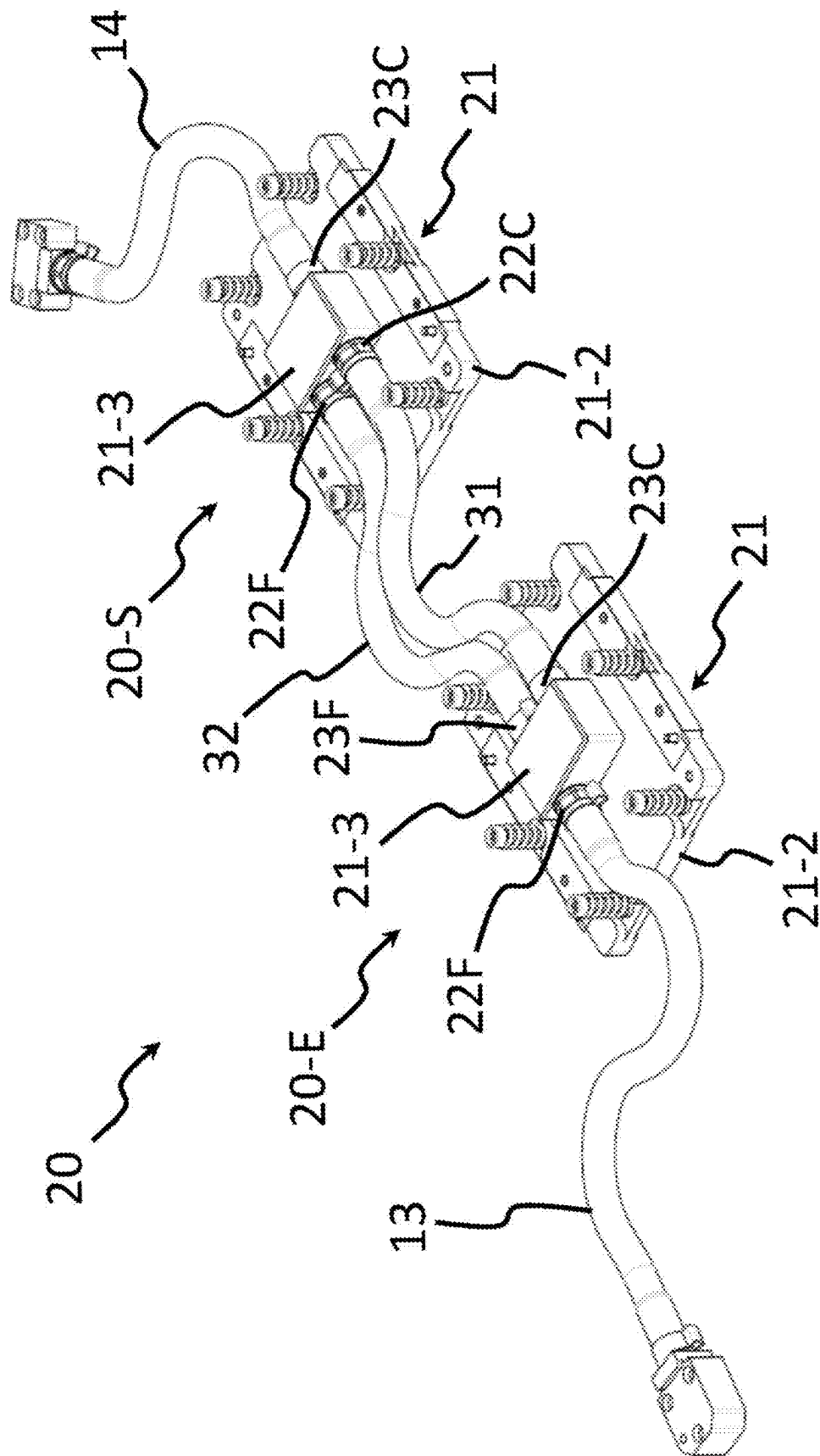


FIGURE 4

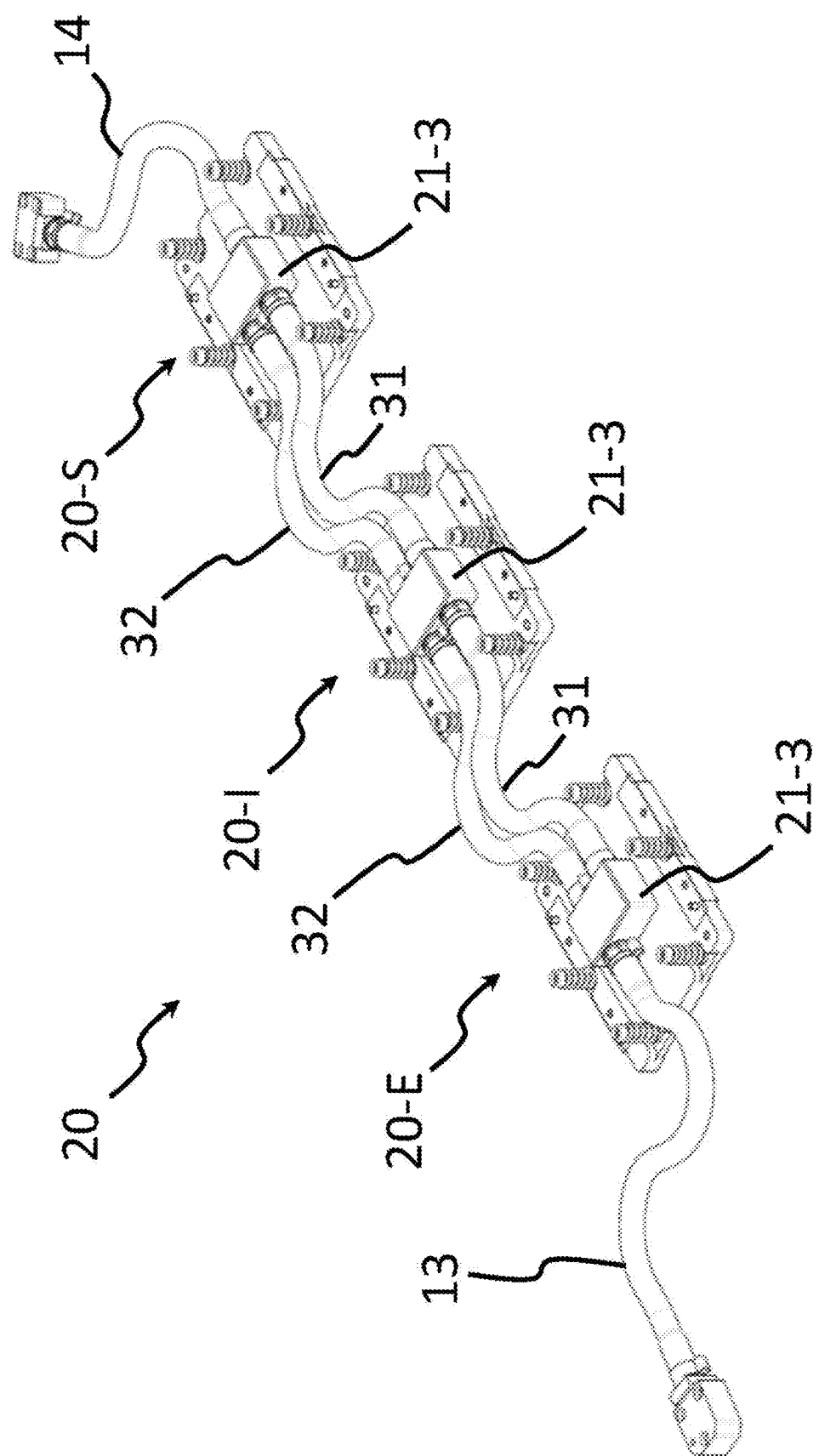


FIGURE 5

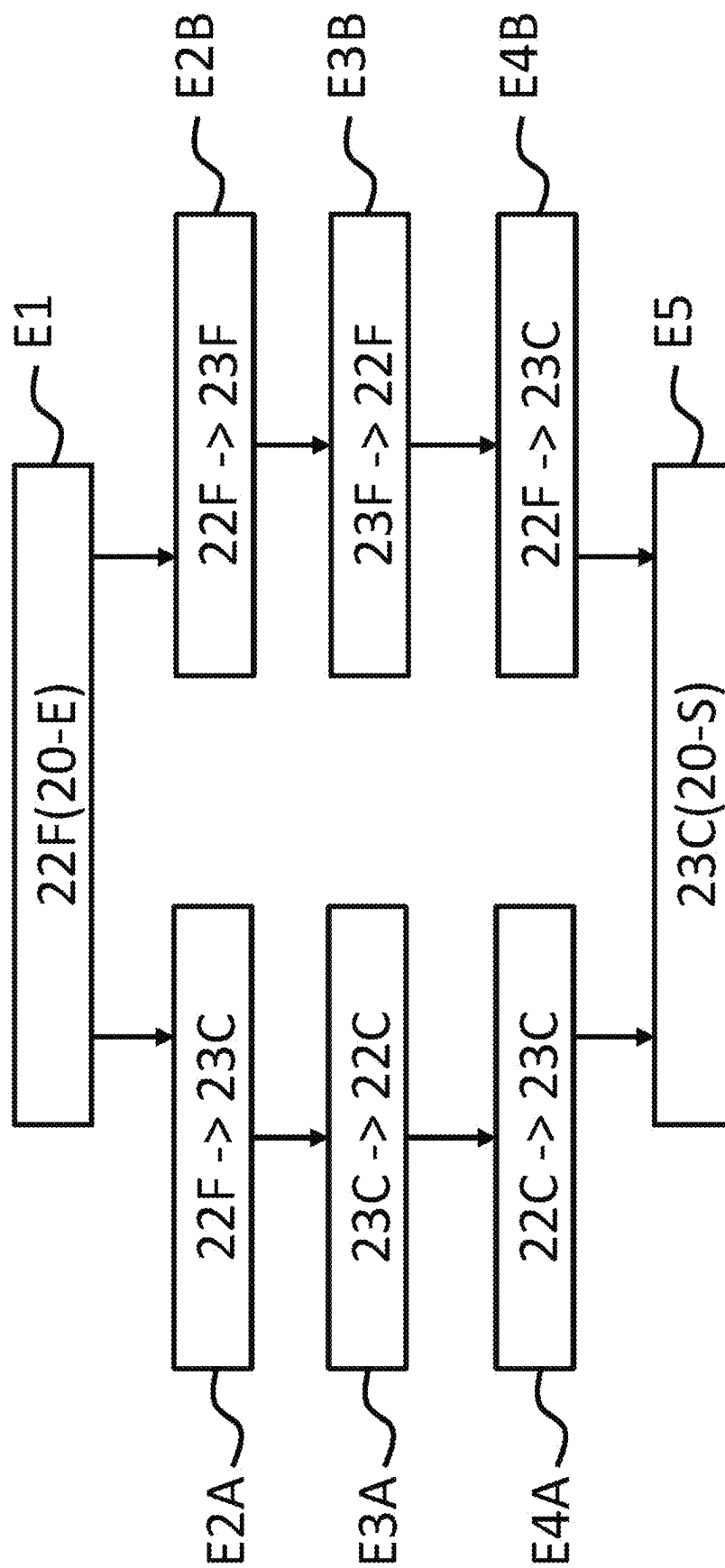


FIGURE 6

1

OUTLET HEAT SINK FOR COOLING SYSTEM OF A SUPERCOMPUTER ELECTRONIC BOARD

This application claims priority to European Patent Appli- 5
cation Number 22306292.8, filed 31 Aug. 2022, the speci-
fication of which is hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

Technical Field of the Invention

At least one embodiment of the invention relates to the 10
field of cooling computer processors and more particularly
relates to a liquid cooling system of an electronic board 15
comprising at least one computing processor.

Description of the Related Art

Server computing blades generally comprise electronic 20
boards, especially processors, which emit heat and therefore
need to be cooled. Initially, these electronic boards were
cooled by air. However, in the case of rack-mountable
servers, air circulation is not sufficient to cool the electronic 25
boards. To solve this problem, fluidic cooling systems have
been developed.

Among the well-known electronic board cooling solu- 30
tions, the company IBM® has developed a dual processor
electronic board cooling system for supercomputers. Such a
system comprises copper pipes containing a heat transfer
fluid forming a liquid loop. However, the rigidity of copper
pipes does not allow quick disassembly of a single proces- 35
sor. In addition, the liquid loop ensures cooling of the
processors only. The rest of the dissipated power is cooled
by convection in the air, which is not optimal for the Power
Usage Effectiveness (PUE) of the supercomputer.

Other electronic board manufacturers provide liquid cool- 40
ing systems dedicated solely to processors. These incorpo-
rate a heat sink, an integrated pump and a heat exchanger for
extracting calories. Of course, such devices are not compat-
ible with larger-scale systems such as supercomputers due to
overall size reasons.

The applicant has further developed a mixed cooling 45
system in which electronic boards are cooled by means of a
cold aluminum plate in which a heat transfer liquid circula-
tes, the cold plate being interfaced with all the low and
medium electronic components to be cooled, that is all the
components of an electronic board excluding the high elec-
tronic components (typically the processors and memory 50
modules). Processors are the most restrictive components to
cool (because they have to remain accessible) and also the
most dissipating. They cannot be interfaced with the cold
plate directly. In order to ensure thermal contact with the
processors, each of them is interfaced with an intermediate
heat sink with heat pipes. It is a phase change, two-phase 55
heat sink (usually referred to as “Heat Spreader CPU”),
which conducts heat from the processors to the cold plate.
The part of the heat pipes in contact with the copper slab that
interfaces the processor constitutes the evaporator, while
that in contact with the interface with the cold plate consti- 60
tutes the condenser. This solution is described in more detail
in European patent applications EP 2770809 and EP
2770810 belonging to the applicant.

This processor cooling solution using heat pipes also has
drawbacks, and in particular:

thermal limits specific to heat pipes: the flow inside a heat
pipe is two-phase and is governed by five limits in

2

terms of flow, viscosity, boiling, driving, capillarity,
which depend on the dimensions of the heat pipes and
the power to discharge.

mechanical limits: the increase in the number of heat
pipes results in an increase in the rigidity of the
assembly. However, in order to compensate for toler-
ances, it is essential that the heat sink deforms suffi-
ciently.

a high number of removable thermal interfaces: thermal
contacts should be ensured between the cold plate and
the heat sink on the one hand, and between the heat sink
and the processor on the other hand. In the current
solution implemented by the applicant, these thermal
contacts are guaranteed by the use of conductive
grease, which has the drawback of generating a high
temperature gradient.

the complexity and cost of the overall solution: the
solid/fluid exchange surfaces are currently at the cold
plate. The complexity of this in terms of heat sinks
considerably increases machining in the cold plate and
hence the manufacturing cost.

To solve the aforementioned defects and drawbacks, the
applicant developed a cooling system for an electronic board
that is fully liquid, comprising a cold plate supplied with
heat transfer fluid via a water path internal to the cold plate,
and a plurality of heat sinks connected as flexible connec-
tions with the water path internal to the cold plate. These
flexible connections form a flexible network coupled to the
network of rigid channels of the cold plate supplied with
heat transfer fluid (typically glycol water), thus allowing the
heat sinks to be supplied with heat transfer fluid. This
network is coupled with the cold plate system allowing
cooling of the rest of the electronic board. In order to
optimize the cooling of the processor as much as possible,
the heat transfer fluid must be brought as close as possible
to the heat source. To this end, the applicant specifically
developed a single-phase heat sink comprising a cooling
block of heat-conducting material that can be supplied with
heat transfer fluid. The heat source constituted by the
processor will therefore be directly interfaced with this
cooling block (also referred to in the examples by the term
“waterblock”) supplied with heat transfer fluid taken from
the cold plate. Such an integrated cooling system meets
restrictions of thermal efficiency, mechanical compactness
and maintainability necessary for this type of high-perfor-
mance computing equipment. This solution is described in
detail in document WO2019115963.

In a known manner, multi-processor calculators require as
many cooling blocks as processors.

For a given chain of components, in a known solution
illustrated in FIG. 1, the heat exchangers **100** are connected
in series, that is the inlet of the first exchanger **100** is
connected to a liquid inlet via an inlet hose **101**, its outlet
is connected to the inlet of the next exchanger **100** via a
connecting hose **102** and so on up to the outlet of the last
exchanger **100** which is connected to a liquid outlet via an
outlet hose **103**. As the exchangers are in series, the tem-
perature of the liquid increases as it approaches the liquid
outlet (the liquid is warmer when it arrives in the last
exchanger as it has already been used to cool the previous
exchangers) and therefore the components located at the end
of the chain are less well cooled than those at the beginning
of the chain. In addition, this solution makes component
maintenance complicated, as all exchangers in flexible con-
nection must be removed to change a component in the
chain, particularly beyond two exchangers in series.

In order to at least partially overcome these drawbacks, it is known to fit the exchangers **100** in parallel and no longer in series, as illustrated in FIG. 2. Thus, each processor is cooled independently of the others and each cooling block can be easily changed in the event of a problem. However, this solution has the drawback of requiring as many connectors (not represented for the sake of clarity) and inlet hoses **104** as there are processors and as many connectors (not represented for the sake of clarity) and outlet hoses **105** as there are processors. However, the connectors and inlet hoses **104** and the connectors and outlet hoses **105** require a significant amount of space to be connected to the cold plate. However, in multi-processor calculators, the lack of space can prove to be an obstacle to installing such a solution.

It would therefore be advantageous to provide a simple, reliable and effective solution to at least partially remedy these drawbacks.

BRIEF SUMMARY OF THE INVENTION

One purpose of at least one embodiment of the invention is therefore to provide a solution for cooling an electronic board, in particular a supercomputer. For this purpose, at least one embodiment of the invention defines several types of complementary heat sinks each having a specific function as well as a cooling system comprising such heat sinks.

Thus, one or more embodiments of the invention firstly relates to an inlet heat sink for a liquid cooling system for an electronic board, the board including at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the heat sink, the cold plate being dimensioned to cover at least partly the electronic board, in particular the support of the at least one electronic component, where applicable, and including a cooling circuit comprising channels inside which a so-called "cold" heat transfer fluid for supplying at least the heat sink circulates, and a discharge circuit comprising channels inside which a so-called "hot" heat transfer fluid that has heated up through the heat sink flows, the heat sink comprising a cooling block of heat-conducting material comprising a lower part constituting the main heat exchange zone and an upper part to be connected to the cold plate, the heat sink including:

- a so-called "cold" inlet connector fluidly connected to the upper part of the cooling block and which is to receive an inlet tube of the cooling circuit so that the cooling block is supplied with the heat transfer fluid from the cold plate,

- a so-called "hot" outlet connector,

- a so-called "cold" outlet connector,

the upper part of the cooling block being configured to divide the heat transfer fluid flow entering through the cold inlet connector into a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and a second flow, oriented directly towards the cold outlet connector.

The inlet heat sink according to one or more embodiments of the invention thus makes it possible to keep part of the cold flow coming from the cold plate to direct it to another heat sink connected to its outlet, thus improving cooling of the electronic component(s) (processor or other) located downstream in the direction of circulation of the heat transfer fluid. The inlet heat sink according to at least one

embodiment of the invention also makes it possible to meet large-scale manufacturing restrictions, due to its simplicity of manufacture.

At least one embodiment of the invention also relates to an outlet heat sink for a liquid cooling system for an electronic board, the board including at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the heat sink, the cold plate being dimensioned to cover at least partly the electronic board, in particular the support for the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called "cold" heat transfer fluid circulates, for supplying the heat sink, and a discharge circuit comprising channels inside which a so-called "hot" heat transfer fluid circulates, having heated up through the heat sink, the heat sink comprising a cooling block made of heat-conducting material comprising a lower part constituting the main heat exchange zone and an upper part intended to be connected to the cold plate, the heat sink including:

- a so-called "cold" inlet connector fluidly connected to the upper part of the cooling block and which is to receive a connecting tube of the cooling circuit so that the cooling block is supplied with the heat transfer fluid from the cold plate (through at least one heat sink),

- a so-called "hot" inlet connector fluidly connected to the upper part of the cooling block and which is to receive a connecting tube of a flow of heated-up heat transfer fluid,

- a so-called "hot" outlet connector,

the upper part of the cooling block being configured, on the one hand, to convey the flow of heat transfer fluid entering through the cold inlet connector to the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the outlet connector, and on the other hand, to convey the flow of heated-up heat transfer fluid received on the hot inlet connector directly to the outlet connector where it fuses with the heated-up flow in the lower part of the cooling block.

The outlet heat sink according to at least one embodiment of the invention thus makes it possible both to receive a flow of cold heat transfer fluid and to fuse hot flows coming from both the lower part of the cooling block and an upstream heat sink connected to the hot inlet of the outlet heat sink, thus improving cooling of the electronic component(s) associated with the outlet heat sink while allowing circulation of the heat transfer fluid. The outlet heat sink according to one or more embodiments of the invention also makes it possible to meet large-scale manufacturing restrictions, due to its simplicity of manufacture.

Advantageously, an inlet heat sink may be connected in pairs with an outlet heat sink to form an assembly of heat sinks according to at least one embodiment of the invention.

At least one embodiment of the invention also relates to an interlayer heat sink for a liquid cooling system for an electronic board, the board comprising at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the heat sink, the cold plate being dimensioned to cover at least partly the electronic board, in particular the support for the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called "cold" heat transfer fluid circulates, for supplying at least the heat sink, and a discharge circuit comprising channels inside which a so-called "hot" heat transfer fluid circulates, which

5

has heated up through the heat sink, the heat sink comprising a cooling block made of heat-conducting material comprising a lower part constituting the main heat exchange zone and an upper part intended to be connected to the cold plate, the heat sink wherein:

a so-called “cold” inlet connector fluidly connected to the upper part of the cooling block and which is to receive an inlet tube of the cooling circuit so that the cooling block is supplied with the heat transfer fluid from the cold plate,

a so-called “hot” inlet connector fluidly connected to the upper part of the cooling block and which is to receive a connecting tube of a flow of heated-up heat transfer fluid,

a so-called “hot” outlet connector,

a so-called “cold” outlet connector,

the upper part of the cooling block being configured to: divide the flow of heat transfer fluid entering through the cold inlet connector into a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and a second flow, oriented directly towards the cold outlet connector,

convey the flow of heated-up heat transfer fluid received on the hot inlet connector directly to the hot outlet connector where it fuses with the heated-up flow in the lower part of the cooling block.

Thus, in at least one embodiment, one or more interlayer heat sinks can be connected between an inlet heat sink and an outlet heat sink of a same assembly of heat sinks in order to cool more electronic components on a same line while allowing conveying of a cold heat transfer fluid flow to the downstream heat sinks of the assembly.

At least one embodiment of the invention also relates to a liquid cooling system for an electronic board, the board comprising at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate and at least one assembly of heat sinks, the cold plate being for receiving the assembly of heat sinks, being dimensioned to cover at least partly the electronic board, in particular the support of the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called “cold” heat transfer fluid circulates, for supplying at least one of the heat sinks, and a discharge circuit comprising channels in which a so-called “hot” heat transfer fluid is heated up through at least one of the heat sinks, each heat sink comprising a main heat exchange zone capable of bearing against one of the processors, each assembly of heat sinks comprising an inlet heat sink as set forth above and an outlet heat sink as set forth above, the inlet connector of the inlet heat sink being fluidly connected to the cooling circuit through an inlet tube, the cold outlet connector of the inlet heat sink being fluidly connected to the cold inlet connector of the outlet heat sink, the hot outlet connector of the inlet heat sink being fluidly connected to the hot inlet connector of the outlet heat sink, the outlet connector of the outlet heat sink being connected to the discharge circuit through an outlet tube.

The system according to one or more embodiment of the invention thus makes it possible to effectively cool the electronic board by distributing the cold heat transfer fluid between the inlet heat sink and the outlet heat sink of a same assembly of heat sinks. The cooling system according to at least one embodiment of the invention makes it possible to guarantee cooling of the electronic board components, by liquid means regardless of the nature of the electronic

6

components: the low and medium components through the cold plate and the so-called high components such as processors and memory modules through the heat sinks, in particular single-phase, with cooling block. The cooling system according to at least one embodiment of the invention, due to its structure with cooling blocks connected as flexible connections with the cooling circuit of the cold plate allows great flexibility for disassembling the processors and guarantees reliable connections after a large number of disassembly operation cycles. A 10 bar tightness is guaranteed by the cooling system according to one or more embodiments of the invention, including after several disassembly operations. The cooling system according to at least one embodiment of the invention, due to its structure with cooling blocks connected as flexible connections with the cooling circuit of the cold plate, allows implantation of a plurality of heat sinks in series or in parallel, for example six in number.

Advantageously, in at least one embodiment, the hot outlet connector of the inlet heat sink is connected to the hot inlet connector of the outlet heat sink through a first flexible and/or rectilinear connecting tube and the cold outlet connector of the inlet heat sink is connected to the cold inlet connector of the outlet heat sink through a second flexible and/or rectilinear connecting tube.

In one or more embodiments, when the first connecting tube and the second connecting tube are rectilinear, they are arranged parallel to each other in order to optimize space and thereby reduce volume of the system.

In at least one embodiment, the at least one assembly of heat sinks comprises an interlayer heat sink as set forth above, fluidly connected on the one hand to the hot outlet connector and the cold outlet connector of the inlet heat sink and on the other hand to the hot inlet connector and the cold inlet connector of the outlet heat sink.

According to one or more embodiments of the invention, the system may comprise a plurality of assemblies of heat sinks, preferably arranged parallel to each other in order to optimize space and thus reduce volume of the system.

Advantageously, in at least one embodiment, the inlet tube of each inlet heat sink is flexible in order to facilitate assembly by flexible connections with the cold plate.

Advantageously, in at least one embodiment, the outlet tube of each outlet heat sink is flexible in order to facilitate assembly by flexible connections with the cold plate.

Further advantageously, in at least one embodiment the at least one interlayer heat sink is connected to the inlet heat sink and/or the outlet heat sink via flexible and/or rectilinear connecting tubes.

At least one embodiment of the invention also relates to a method for separating a flow of “cold” heat transfer fluid in an inlet heat sink as set forth above for a liquid cooling system for an electronic board, the board comprising at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the heat sink, the cold plate being dimensioned to cover at least partly the electronic board, in particular the support of the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called “cold” heat transfer fluid, for supplying at least the heat sink circulates, and a discharge circuit comprising channels inside which a so-called “hot” heat transfer fluid has heated up through the heat sink, the method comprising the steps of supplying the cold inlet connector with the heat transfer fluid from the cold plate and dividing the heat transfer fluid flow entering through the cold inlet

connector into a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and a second flow, oriented directly to the cold outlet connector.

At least one embodiment of the invention also relates to a method for melting a heat transfer fluid stream into an outlet heat sink as set forth above for a liquid cooling system for an electronic board, the board comprising at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the heat sink, the cold plate being dimensioned to cover at least partly the electronic board, in particular the support of the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called "cold" heat transfer fluid, for supplying the heat sink, and a discharge circuit comprising channels in which a so-called "hot" heat transfer fluid is heated up through the heat sink, the method comprising the steps of supplying the cold inlet connector with the cold heat transfer fluid, supplying the hot inlet connector with a flow of heated-up heat transfer fluid, conveying the heat transfer fluid flow entering through the cold inlet connector to the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the outlet connector and conveying the flow of heated-up heat transfer fluid received on the hot inlet connector directly to the outlet connector where it fuses with the heated-up flow in the lower part of the cooling block where it fuses with the heated-up flow in the lower part of the cooling block.

At least one embodiment of the invention also relates to a method for managing a flow of heat transfer fluid in an interlayer heat sink as set forth above for a liquid cooling system for an electronic board, the electronic board comprising at least one electronic component, for example a processor, a voltage regulator or a memory module, for example mounted to a support, the system comprising a cold plate for receiving the interlayer heat sink, the cold plate being dimensioned to cover at least partly the electronic board especially the support of the at least one electronic component, where applicable, and comprising a cooling circuit comprising channels inside which a so-called "cold" heat transfer fluid, for supplying at least the heat sink, and a discharge circuit comprising channels inside which a so-called "hot" heat transfer fluid is heated up through the heat sink, the method comprising the steps of:

supplying the cold inlet connector with the heat transfer fluid from the cold plate,

dividing the flow of heat transfer fluid entering through the cold inlet connector into a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and a second flow, oriented directly to the cold outlet connector,

supplying the hot inlet connector with a flow of heated-up heat transfer fluid,

conveying of the heated-up heat transfer fluid flow received on the hot inlet connector directly to the hot outlet connector where it fuses with the heated-up flow in the lower part of the cooling block.

The heat sink of the cooling system according to one or more embodiments of the invention must furthermore itself respect a space restriction defined by the size of the components (processors or others) and by the space available on the cold plates to judiciously distribute the cooling fluid.

BRIEF DESCRIPTION OF THE DRAWINGS

Further characteristics and advantages of the invention will further appear upon reading the description that follows. This is purely illustrative and should be read in conjunction with the appended drawings in which:

FIG. 1 schematically illustrates one embodiment of a system of prior art in which the heat sinks are mounted in series.

FIG. 2 schematically illustrates one embodiment of a system of prior art in which the heat sinks are mounted in parallel.

FIG. 3 schematically illustrates a system according to one or more embodiments of the invention.

FIG. 4 schematically illustrates, partially, an assembly of heat sinks of the system of FIG. 3, according to one or more embodiments of the invention.

FIG. 5 schematically illustrates, partially, an assembly of heat sinks comprising an interlayer heat sink, according to one or more embodiments of the invention.

FIG. 6 schematically illustrates a method according to one or more embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 3 illustrates an example of the system 1 according to one or more embodiments of the invention.

The system 1, in at least one embodiment, allows liquid cooling of an electronic board 5 (partially visible in transparency) and for this purpose comprises a cold plate 10 and assemblies 20 of heat sinks 20-E, 20-S, one of which is represented separately (and partially) in FIG. 4 by way of one or more embodiments.

In this example, in one or more embodiments, the electronic board 5 comprises six processors mounted to a support (not visible) of the electronic board 5. The electronic board 5 comprises a plurality of electronic components attached to the support, broken down for example into low and medium electronic components and high electronic components comprising the processors.

For the purposes of one or more embodiments of the invention, by "low and medium electronic components", it is meant any component of the electronic board 5 having a height below a threshold height, below which it can be cooled by the cooling circuit of the cold plate 10. For low and medium electronic components, in at least one embodiment, these are usually power supplies or voltage regulators.

For the purposes of one or more embodiments of the invention, by "high electronic components", it is meant any component of the electronic board 5 which is not cooled by the cooling circuit of the cold plate 10 because they have to remain accessible without disassembly of the cold plate 10. These are electronic components that require fast maintenance such as processors or memory modules.

In the example in FIG. 3, by way of at least one embodiment, the system 1 comprises, in addition to the cold plate 10, three assemblies 20 of heat sinks 20-E, 20-S arranged parallel to each other and each comprising an inlet heat sink 20-E and an outlet heat sink 20-S.

The cold plate 10 is dimensioned to cover the support of the processors. In this example, by way of at least one embodiment, the cold plate 10 has external dimensions substantially equal to those of the support of the electronic board 5 so that it covers the entire support and the low and medium electronic components.

The cold plate 10 comprises a cooling circuit 10R and a discharge circuit 10E. The cooling circuit 10R comprises channels, preferably rigid, inside which a so-called “cold” heat transfer fluid circulates for supplying the inlet heat sinks. The discharge circuit 10E comprises channels inside which a so-called “hot” heat transfer fluid circulates, which has heated up through the heat sinks 20-E, 20-S.

The supply with cold heat transfer fluid to the cold plate 10 is carried out via an inlet connector 10in and the outlet of the heated heat transfer fluid from the cold plate 10 is carried out via an outlet connector 10out.

The heat sinks 20-E, 20-S can be of the single-phase type.

Each heat sink 20-E, 20-S comprises a main heat exchange zone able to bear against a high electronic component, especially one of the processors, or a low electronic component, especially the voltage regulators and/or memory modules.

Referring to FIG. 4, by way of one or more embodiments, each heat sink 20-E, 20-S comprises a cooling block 21, at least one inlet connector 22C, 22F, and at least one outlet connector 23C, 23F (depending on the type of heat sink 20-E, 20-S).

The cooling block 21 is made of a heat-conducting material such as, for example, aluminum or aluminum alloy.

The cooling block 21 comprises a lower part 21-1 constituting the main heat exchange zone, a so-called “distribution” (optional) intermediate part 21-2 and an upper part 21-3 connected to the cold plate 10.

In at least one embodiment, the cooling block 21 comprises a through slot located in the intermediate part for homogeneously distributing the flow of the heat transfer fluid from the upper part in the form of a jet centered towards a support plate located in the main heat exchange zone and comprising a hollow central part with a zone provided with fins or pins to generate turbulence in the flow regime of the heat transfer fluid and a zone peripheral to the zone provided with fins or pins to collect heat transfer fluid and direct it towards the outlet connector. In at least one embodiment, the slot is thin. Thus, by way of one or more embodiments, in this configuration, the heat transfer fluid, when it arrives in the cooling block 21, passes through the thin slot, which allows it to be equally distributed in the exchange surface through the fins. The shape of this distributor and its distance from the fins are designed to obtain the largest possible Reynolds number in the available overall space.

Inlet Heat Sink 20-E

Still with reference to FIGS. 3 and 4, by way of one or more embodiments, each inlet heat sink 20-E comprises a so-called “cold” inlet connector 22F, a so-called “hot” outlet connector 23C and a so-called “cold” outlet connector 23F.

The cold inlet connector 22F is fluidly connected to the upper part of the cooling block 21 and receives an inlet, preferably flexible, tube 13 connected to the cooling circuit 10R to supply the cooling block 21 with the heat transfer fluid from the cold plate 10.

The upper part of the cooling block 21 is configured to divide the heat transfer fluid flow entering through the cold inlet connector 22F into a first flow and a second flow. In at least one embodiment, this division is carried out by implementing a Tickelman loop, known per se, at the cooling block 21.

The first flow is directed towards the lower part of the cooling block 21 so as to collect heat generated by the electronic board 5 by being conveyed to the hot outlet connector 23C.

The second flow is directed directly to the cold outlet connector 23F.

Outlet Heat Sink 20-S

Each outlet heat sink 20-S comprises a so-called “hot” inlet connector 22C, a so-called “cold” inlet connector 22F and a so-called “hot” outlet connector 23C.

The hot inlet connector 22C is fluidly connected to the upper part of the cooling block 21 and receives a first, preferably flexible, connecting tube 31 of a heat transfer fluid flow heated up by the inlet heat sink 20-E, connected to the hot outlet connector 23C of the inlet heat sink 20-E.

The cold inlet connector 22F is fluidly connected to the upper part of the cooling block 21 and receives a second, preferably flexible, connecting tube 32, connected to the cold outlet connector of the inlet heat sink 20-E, so that the cooling block 21 is supplied with the heat transfer fluid from the cold plate 10 through the inlet heat sink 20-E, especially from the cold outlet connector 23F of the inlet heat sink 20-E.

The upper part of the cooling block 21 is configured to, on the one hand, convey the flow of heat transfer fluid entering through the cold inlet connector 22F to the lower part of the cooling block 21 so as to collect heat generated by the electronic board 5 by being conveyed to the hot outlet connector 23C, and, on the other hand, to convey the heated-up heat transfer fluid received on the hot inlet connector 22C directly to the hot outlet connector 23C. In at least one embodiment, this fusion is achieved by implementing a Tickelman loop, known per se, at the cooling block 21.

The hot outlet connector 23C of the outlet heat sink 20-S is connected to the discharge circuit of the cold plate through an outlet tube 14.

In at least one embodiment, illustrated in FIG. 5, each assembly of heat sinks further comprises an interlayer heat sink 20-I, fluidly connected on the one hand to the hot outlet connector 23C and the cold outlet connector 23F of the inlet heat sink 20-E and on the other hand to the hot inlet connector 22C and the cold inlet connector 22F of the outlet heat sink 20-S. This configuration allows cooling a matrix of 3x3 (i.e. 9) processors (or more) simultaneously.

The interlayer heat sink 20-I comprises a so-called “cold” inlet connector 22F to be fluidly connected to the cold outlet connector 23F of the inlet heat sink 20-E and fluidly connected to the upper part 21-3 of the cooling block 21 so that the cooling block 21 is supplied with the cold heat transfer fluid. The interlayer heat sink 20-I comprises a so-called “hot” inlet connector 22C to be fluidly connected to the hot outlet connector 23C of the inlet heat sink 20-E and fluidly connected to the upper part 21-3 of the cooling block 21. The interlayer heat sink 20-I comprises a so-called “hot” outlet connector 23C to be fluidly connected to the hot inlet connector 22C of the outlet heat sink 20-S. The interlayer heat sink 20-I comprises a so-called “cold” outlet connector 23F to be fluidly connected to the cold inlet connector 22F of the outlet heat sink 20-S. The upper part 21-3 of the cooling block 21 is configured to divide the flow of heat transfer fluid entering through the cold inlet connector 22F into a first flow, which is oriented towards the lower part 21-1 of the cooling block 21 so as to collect heat generated by the electronic board 5 by being conveyed to the hot outlet connector 23C, and a second flow, oriented directly towards the cold outlet connector 23F, and convey the received heated heat transfer fluid flow on the hot inlet connector 22C directly to the hot outlet connector 23C.

11

In at least one embodiment, the system **1** could comprise more or less than three assemblies of heat sinks and each assembly of heat sinks could comprise, in addition to an inlet heat sink **20-E** and an outlet heat sink **20-S**, one or more interlayer heat sinks **20-I**, connected in series.

Exemplary Implementation

The heat transfer fluid circulating in the cooling circuit of the cold plate, before entering the inlet heat sinks, is referred to as “cold” fluid. The heat transfer fluid circulating in the cold plate discharge circuit after passing through the heat sink assemblies is referred to as “hot” fluid.

Thus, with reference to FIG. 6, by way of one or more embodiments, in a step **E1**, the cold heat transfer fluid is conveyed by the cooling circuit to the cold inlet connector **22F** of each inlet heat sink **20-E** in order to supply it.

Each inlet heat sink **20-E** divides the flow of cold heat transfer fluid entering through the cold inlet connector **22F** into a first flow and a second flow, substantially equal in flow rate, the flow rate being about half the flow rate of the incoming flow.

The first flow is oriented from the cold inlet connector **22F** to the lower part **21-3** of the cooling block **21** in a step **E2A** so as to collect heat generated by the electronic board **5** and then be conveyed to the hot outlet connector **23C**.

The second flow is oriented in a step **E2B** directly to the cold outlet connector **23F** without passing through the lower part **21-3** of the cooling block **21** and therefore without undergoing heating up by the electronic board **5**.

In a step **E3A**, the first heated-up flow is conveyed from the hot outlet connector **23C** of the inlet heat sink **20-E** to the hot inlet connector **22C** of the outlet heat sink **20-S** via the first connecting tube **31**.

In a step **E3B**, the second flow, comprised of cold heat transfer fluid, is conveyed at the inlet of the cold inlet connector **22F** of the outlet heat sink **20-S**.

In a step **E4A**, the flow of heated-up heat transfer fluid is transferred directly from the hot inlet connector **22C** of the outlet heat sink **20-S** to the hot outlet connector **23C** of the outlet heat sink **20-S**.

In a step **E4B**, the second flow is conveyed to the lower part **21-3** of the cooling block **21** of the outlet heat sink **20-S** so as to collect heat generated by the electronic board **5**, then to the hot outlet connector **23C** of the outlet heat sink **20-S** where it mixes, in a step **E5**, with the heated-up heat transfer fluid flow provided by step **E4A**, so as to regain the initial flow rate of the inlet flow of step **E1**.

The one or more embodiments of the invention makes it possible to have the assemblies **20** of heat sinks **20-E**, **20-S** in parallel by reducing the number of inlet tubes **13** and outlet tubes **14** while allowing easy maintenance.

The invention claimed is:

1. An outlet heat sink for a liquid cooling system of an electronic board, said electronic board comprising at least one electronic component, said liquid cooling system comprising a cold plate that receives said outlet heat sink, said cold plate being dimensioned to cover at least partly said electronic board and comprising a cooling circuit comprising channels inside which a cold heat transfer fluid circulates, to supply said outlet heat sink, and a discharge circuit comprising channels inside which a hot heat transfer fluid circulates, which has been heated up through the outlet heat sink, wherein the outlet heat sink comprises:

a cooling block comprising a lower part constituting a main heat exchange zone and an upper part to be connected to the cold plate,

12

a cold inlet connector fluidly connected to the upper part of said cooling block and which is to receive a connecting tube of the cooling circuit such that said cooling block is supplied by heat transfer fluid from the cold plate,

a hot inlet connector fluidly connected to the upper part of said cooling block and which is to receive a connecting tube of a flow of heated-up heat transfer fluid,

a hot outlet connector, the upper part of the cooling block being configured to convey, on one hand, a flow of heat transfer fluid entering through the cold inlet connector to the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and, on another hand, to convey the flow of heated-up heat transfer fluid received on the hot inlet connector directly to the hot outlet connector.

2. The outlet heat sink according to claim 1, wherein the at least one electronic component is a processor, a voltage regulator, or a memory module.

3. The outlet heat sink according to claim 1, wherein the cooling block is made of a heat-conducting material.

4. A liquid cooling system for an electronic board, said electronic board comprising at least one electronic component, said liquid cooling system comprising:

a cold plate and at least one assembly of heat sinks,

wherein the cold plate is configured to receive said at least one assembly of heat sinks, being dimensioned to at least partly cover the electronic board and wherein the cold plate comprises

a cooling circuit comprising channels inside which a cold heat transfer fluid circulates, to supply at least one heat sink of the at least one assembly of heat sinks, and

a discharge circuit comprising channels inside which a hot heat transfer fluid has heated up through at least one heat sink of the at least one assembly of heat sinks,

wherein each heat sink of the at least one assembly of heat sinks comprises a main heat exchange zone capable of bearing against one electronic component of the at least one electronic component,

wherein each assembly of heat sinks of the at least one assembly of heat sink comprises an outlet heat sink and an inlet heat sink, comprising

a cooling block comprising a lower part constituting the main heat exchange zone and an upper part to be connected to the cold plate,

a cold inlet connector fluidly connected to the upper part of said cooling block and receiving an inlet tube of the cooling circuit such that said cooling block is supplied by heat transfer fluid from the cold plate,

a hot outlet connector,

a cold outlet connector,

wherein the upper part of the cooling block is configured to divide a flow of heat transfer fluid entering through the cold inlet connector into a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and a second flow, oriented directly towards the cold outlet connector, the cold outlet connector of the inlet heat sink being fluidly connected to the cold inlet connector of the outlet heat sink, the hot outlet connector of the inlet heat sink being

13

fluidly connected to a hot inlet connector of the outlet heat sink, the hot outlet connector of the outlet heat sink being connected to the discharge circuit by an outlet tube.

5. The liquid cooling system according to claim 4, wherein

the hot outlet connector of the inlet heat sink is connected to the hot inlet connector of the outlet heat sink through one or more of

a first flexible connecting tube, and
a first rectilinear connecting tube, and

the cold outlet connector of the inlet heat sink is connected to the cold inlet connector of the outlet heat sink through one or more of

a second flexible connecting tube, and
a second rectilinear connecting tube.

6. The liquid cooling system according to claim 4, wherein the at least one assembly of heat sinks comprises an interlayer heat sink comprising said cooling block comprising said lower part constituting the main heat exchange zone and said upper part to be connected to the cold plate, wherein the interlayer heat sink comprises

said cold inlet connector fluidly connected to the cold outlet connector of the inlet heat sink and fluidly connected to the upper part of said cooling block such that said cooling block is supplied by the cold heat transfer fluid,

said hot inlet connector fluidly connected to the hot outlet connector of the inlet heat sink and fluidly connected to the upper part of said cooling block,

said hot outlet connector fluidly connected to the hot inlet connector of the outlet heat sink,

said cold outlet connector fluidly connected to the cold inlet connector of the outlet heat sink, the upper part of the cooling block being configured to

divide the flow of heat transfer fluid entering through the cold inlet connector into

a first flow, which is oriented towards the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and

a second flow, oriented directly towards the cold outlet connector,

convey a flow of heated-up heat transfer fluid received on the hot inlet connector directly to the hot outlet connector.

14

7. The liquid cooling system according to claim 4, wherein the at least one assembly of heat sinks comprises a plurality of assemblies of heat sinks arranged in parallel to each other.

8. The liquid cooling system according to claim 4, wherein the inlet tube of each inlet heat sink is flexible.

9. The liquid cooling system according to claim 4, wherein the outlet tube of each outlet heat sink is flexible.

10. A method for fusing a flow of heat transfer fluid in an outlet heat sink for a liquid cooling system for an electronic board, said electronic board comprising at least one electronic component, said liquid cooling system comprising a cold plate for receiving said outlet heat sink, said cold plate being dimensioned to cover at least partly the electronic board and comprising a cooling circuit comprising channels inside which a cold heat transfer fluid, for supplying said outlet heat sink, and a discharge circuit comprising channels inside which a hot heat transfer fluid is heated through the outlet heat sink,

wherein the outlet heat sink comprises

a cooling block comprising a lower part constituting a main heat exchange zone and an upper part to be connected to the cold plate,

a cold inlet connector fluidly connected to the upper part of said cooling block and which is to receive a connecting tube of the cooling circuit such that said cooling block is supplied by heat transfer fluid from the cold plate,

a hot inlet connector fluidly connected to the upper part of said cooling block and which is to receive a connecting tube of a flow of heated-up heat transfer fluid,

a hot outlet connector;

the method comprising:

supplying the cold inlet connector with the cold heat transfer fluid,

supplying the hot inlet connector with said flow of heated-up heat transfer fluid,

conveying the heat transfer fluid entering through the cold inlet connector to the lower part of the cooling block so as to collect heat generated by the electronic board by being conveyed to the hot outlet connector, and

conveying the heated-up heat transfer fluid received to the hot inlet connector directly to the hot outlet connector where it fuses with the flow of heated-up transfer fluid in the lower part of the cooling block.

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